Large Language Models for Bayesian Network Parameterization in Software Engineering: Direct Elicitation versus Ranked Nodes

Anonymous Author(s)

Abstract

Software engineering (SE) decision makers need models they can understand and use when data are scarce. Bayesian networks (BNs) make probabilistic relationships explicit and incorporate expert knowledge, but eliciting their conditional probability tables (CPTs) is costly. The Ranked Nodes Method (RNM) generates ordinal CPTs from fewer judgments. Should large language models (LLMs) fill complete CPTs or supply fewer judgments to RNM? We compared complete CPTs for 61 ordinal conditional nodes from seven SE BNs in an exploratory study. We tested structured prompts that clarified conditioning, framed probabilities as hypothetical frequencies, and requested consistency checks. Qualitative retrieval-augmented generation (RAG) supplied short passages on domain relationships, without reference probabilities or numerical parameters. Eight conditions used GPT-6 Astra at medium effort across five replications. With equal node weights, total variation averaged 0.2317 for direct elicitation and 0.2404 for RNM, a 3.76% increase. Because nodes were unevenly distributed across five source studies, we weighted each study equally; direct elicitation still had lower error. Both contrasts passed Holm-adjusted paired t-tests, but not exact sign tests. RNM improved 31 nodes, yet larger deteriorations elsewhere outweighed those gains. Structured prompting modestly helped direct elicitation under node weighting; qualitative RAG demonstrated no benefit. RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, but only 0.61% fewer total tokens. The findings favor direct elicitation as the average numerical baseline in this sample and show why reduced elicitation alone cannot establish a better, cheaper to complete CPT.

CCS Concepts

• **Computing methodologies** → **Bayesian networks**; • **Software and its engineering** → *Software creation and management*.

Keywords

Bayesian networks, large language models, probability elicitation, ranked nodes, software engineering

1 Introduction

Supporting software engineering (SE) decisions faces two main challenges: explaining the basis for a recommendation and obtaining enough relevant data to build a model experts can rely on. Examples include release readiness, quality assessment, and process improvement, where decision makers must judge whether a recommendation's assumptions apply. A review of 108 studies documents the demand for explainability across 23 SE tasks [3]. Training data for machine learning and deep learning may also be scarce, particularly for managerial SE decisions [23]. Explicit assumptions and domain knowledge are therefore valuable.

SAC '27, Gwangju, South Korea
2027.

Bayesian networks (BNs) offer such an approach. A BN makes relationships and uncertainty explicit in a graph and probabilities, incorporating domain knowledge when data are limited [10, 11]. BNs have supported many SE problems, including early defect prediction [10], effort estimation [17], and diagnosis of development-process problems [21]. A systematic mapping of 106 studies identifies software management and quality as prominent application areas [24]. Building a credible BN nevertheless requires substantial knowledge engineering effort.

Constructing conditional probability tables (CPTs) is a major part of that effort. Each row describes a child's probabilities for one combination of its parents' states; adding parents rapidly increases the number of judgments required [11]. Semi-automatic methods reduce this burden by reconstructing a table from fewer expert judgments or parameters. These include the Ranked Nodes Method (RNM) [11], the Weighted Sum Algorithm (WSA) [20], Analytic Hierarchy Process (AHP)-based pairwise comparisons [5], and interpolation methods [1, 2]. Their assumptions can, however, limit fidelity to the intended probabilities [2].

Large language models (LLMs) are now being explored for CPT construction. Nafar et al. elicited numerical BN parameters directly [18]; Gottal and Matthes reconstructed CPTs from statistical summaries extracted from scientific literature [12]; and Cypko et al. used retrieval-augmented generation (RAG) to support expert BN modeling [6]. Should an LLM specify every row, or supply fewer judgments for a semi-automatic method to complete? Any benefit depends on both the judgments and the reconstruction.

We investigated this question using RNM, which applies to variables with ordered states such as Low, Medium, and High [11, 15]. Instead of requesting the entire CPT, we asked the LLM for distributions at selected parent-state combinations, called *anchors*, and for a rule describing how the parents combine. A numerical fitting procedure then adjusted that rule's weights and uncertainty to match the elicited distributions and generated the complete CPT. We compared this approach with direct complete-table elicitation on 61 ordinal conditional nodes from seven SE BNs.

We also examined support for elicitation. Structured prompts made conditioning explicit and asked for probability and consistency checks; qualitative RAG supplied short passages explaining relevant domain relationships. Crossing the two elicitation methods, two prompt designs, and presence or absence of RAG produced eight conditions. We repeated each five times using GPT-6 Astra at medium effort. This study addresses three research questions:

- **RQ1—Reference agreement:** How does LLM-assisted RNM compare with direct elicitation in complete-CPT agreement across the full SE sample?
- **RQ2—Elicitation support:** How do structured prompting and qualitative RAG affect each method's agreement across complete tables?

- **RQ3—Accuracy and effort:** How does RNM trade reference agreement against requested judgments and observed computational effort?

All three RQs include all 61 nodes and every CPT row, including anchors. The seven BNs cover Scrum process assessment [19], Chrome code review [14], ERP data quality [13], software maintenance logs [8], and software strategic indicators [16]. We give nodes equal weight for the main comparison. Because BNs contribute unequal numbers of nodes, we also weight their five source studies equally and report each BN separately. This evaluates the complete tables available for use rather than restricting attention to configurations omitted from RNM prompts.

Section 2 positions the study in related work, Section 3 describes the experiment, and Section 4 reports the results. Section 5 discusses the RQs and implications for SE, Section 6 examines validity threats, and Section 7 concludes the paper.

2 Background and Related Work

This section explains RNM and briefly introduces alternative CPT construction techniques (Section 2.1), then examines LLM approaches for CPT construction and positions our contributions (Section 2.2).

2.1 Ranked Nodes Method

RNM generates CPTs for ordinal variables such as Low, Medium, and High from a compact description of parent influence and uncertainty [11, 15]. Ordered states correspond to intervals on a common $[0, 1]$ scale. For example, three equally spaced states occupy successive thirds of that scale. For each parent-state configuration, a weighted function combines values from the corresponding parent intervals. A normal distribution, truncated to $[0, 1]$ and centered at the function value before truncation, assigns probability mass to the child's state intervals. Averaging over parent values produces the CPT row.

The function, weights, and variance express different judgments. A weighted mean allows stronger parents to compensate for weaker ones; weighted-minimum and weighted-maximum functions emphasize weaker or stronger parent values. Weights adjust parent influence, while variance controls dispersion across child states. Thus experts specify a relationship and uncertainty rather than independently assessing every row. Parameters can also be fitted to selected elicited distributions, as in our study. The same settings generate all rows, reducing elicitation but restricting the relationships representable. Laitila and Virtanen show that RNM can approximate many CPT relationships while remaining sensitive to parameter choices [15].

RNM is one of several semi-automatic approaches. WSA combines elicited distributions through weighted sums [20]; AHP uses pairwise comparisons [5]; interpolation methods, including InterBeta, infer unassessed distributions from selected inputs [1, 2]. These alternatives use different judgments and assumptions, so reduced elicitation alone does not establish numerical fidelity.

2.2 LLMs for CPT construction

LLM approaches differ in elicitation granularity and supplied evidence. Nafar et al. found that requesting one child-state probability at a time lowered median BN KL divergence for seven of eight models, with more queries [18]. Their FullDist baseline requests one conditional row; ours requests the whole CPT. Their result leaves open whether eliciting fewer configurations and reconstructing the rest improves agreement. Gottal and Matthes reconstruct from numerical summaries [12], which constrain probabilities more directly than qualitative RAG. Rahul and Chowdhury's persona-panel preprint evaluates effort and internal consistency, without independent reference-CPT accuracy [22].

Other work evaluates modeling or predictive support. BIRD illustrates neural–symbolic integration [7] through LLM judgments, constrained fitting, and BN inference [9]. Clinical RAG assists BN modelers [6]; AutoElicit evaluates prediction after updating elicited parameter priors [4]. These outcomes complement numerical CPT agreement but cannot establish it.

We connect these research directions by comparing direct elicitation with LLM anchors plus RNM fitting across complete CPTs for 61 nodes from seven SE BNs. We examine structured prompting, qualitative RAG, and token usage, with anchor diagnostics separating elicitation from fitting. Experimental fits use LLM judgments without reference access. Table 1 compares the studies by supplied evidence, evaluation endpoint, and scope.

3 Method

We crossed direct versus RNM elicitation, baseline versus structured prompting, and absence versus presence of qualitative RAG. Each of the eight conditions covered 61 nodes in five replications ($61 \times 8 \times 5 = 2{,}440$ trials). We compared complete outputs against hidden reference CPTs (RQ1), assessed prompting and evidence effects (RQ2), and measured elicitation and token usage (RQ3).

3.1 BNs and reference CPTs

The sample contains seven BNs from five published SE studies (Table 2). Scrum assesses team and process factors affecting sprint and backlog quality. Chrome 40 and 42 concern code review and post-release defects in two software versions. APS concerns ERP data quality for advanced planning and scheduling; the two software-log BNs concern maintenance logs; SESSI concerns software strategic indicators. Versions from the same study form one publication family.

We selected conditional nodes with complete numerical references and interpretable, ordered child and parent states. Binary variables qualified when their meanings supported an order; nominal variables and Boolean gate chains were excluded. The selected CPTs contain 421 rows, with at most 27 rows per node. Scrum uses an RNM-derived reference; the other BNs supply tabulated probabilities. These references represent particular modeling contexts, not universal company probabilities. The supplement describes their versions and preparation.

All RQs include every CPT row, including anchors. Single-parent and fully elicited nodes remain eligible because fitting can change their probabilities. Sampling is availability-based; the endpoint, weighting, and tests followed inspection of earlier results, making this analysis exploratory.

Table 1: LLM-assisted BN construction compared by inputs, evaluation, and scope. Entries distinguish numerical CPT agreement from modeling support and downstream performance; reported errors are not directly comparable. KL denotes Kullback–Leibler divergence.

Study	Construction and supplied evidence	Evaluation material and endpoint	Scope relative to our comparison
Nafar et al. [18] TMLR, 2026	LLM internal knowledge; one conditional row or one child-state probability per query. A separate stage combines estimates with observations.	80 public BNs, eight LLMs: BN/CPT KL. Separate classification evaluation on nine datasets with three LLMs.	Changes granularity within a row; our RNM approach requests selected rows and reconstructs the rest.
Gottal and Matthes [12] Findings ACL, 2026	Reconstruction algorithms and direct LLM CPT output from numerical statistical summaries; separate extraction from publications.	Synthetic BNs and publication/raw-data cases; CPT distances, inference and decision utility, and runtime.	Quantitative summaries constrain construction; our RAG packets supply qualitative context.
Rahul and Chowdhury [22] Preprint, 2026	Persona-conditioned LLM panels provide distributions for parent configurations; trimmed-mean aggregation.	One healthcare-consultation BN; plausibility and internal consistency checks; calls, token estimates, and runtime.	Demonstrates construction feasibility and effort; lacks an independent numerical CPT benchmark.
Feng et al. (BIRD) [9] ICLR, 2025	LLM-generated factors and coarse likelihood judgments feed constrained parameter fitting and BN inference.	Human judgments of relative support and downstream task accuracy.	Establishes a precedent for LLM judgments plus fitting; does not test numerical reference-CPT recovery.
Cypko et al. [6] IJCARS, 2026	Clinical RAG supports expert BN modeling using guidelines and textbooks.	Synthetic retrieval evaluation; modeling, usability, and workload study with four clinicians.	Tests modeling support rather than complete numerical CPT agreement.
This study	Complete CPT elicitation versus LLM anchors plus RNM fitting; baseline/structured prompts, with/without qualitative RAG.	61 ordinal conditional nodes from seven SE BNs; complete-table distances, five replications, tokens, and fitting time.	Compares final CPTs on all rows of all nodes; checks weighting sensitivity and uses anchor fitting as a diagnostic.

Table 2: Evaluated BNs and CPTs. Rows count conditional distributions across the selected nodes; anchors are the subset requested in each RNM condition. Counts are per replication and prompt/evidence combination.

BN and source	Nodes	CPT rows	Anchors
Scrum [19]	25	249	123
Chrome 40 [14]	15	78	57
Chrome 42 [14]	12	63	46
APS [13]	3	8	8
SESSI [16]	1	3	3
Software logs 3 [8]	4	16	12
Software logs 4 [8]	1	4	3
Total	61	421	252

3.2 Inputs and outputs in the eight conditions

Common inputs. Requests supplied an expert role, SE setting, variable definitions, ordered states, assumed parent influence directions, and configurations to evaluate. Answers conditioned on all listed parents and averaged over unspecified circumstances. Graphs, definitions, prompts, and numerical procedures were fixed before collection.

Methods and simplified prompts. Cohesion depends on Self-Organizing and Collaboration; all use Low, Medium, and High states. Its brief describes synchronous work toward shared goals, adapting work, and mutual support. These simplified excerpts illustrate the method-specific requests:

> **Direct:** Estimate probabilities for Low, Medium, and High Cohesion given each listed combination of Self-Organizing and Collaboration. Return all nine distributions in JSON, preserving state order and normalization.
> **RNM:** Return only the five listed anchor distributions. Select a weighted-mean, weighted-minimum, weighted-maximum, or minimum–maximum rule. Do not supply weights, variance, or other rows; these will be fitted offline.

Both methods supplied normalized vectors and an evidence note. Only RNM enforced the shared influence assumptions mathematically. Complete prompts and schemas are in the replication package.

Prompt and evidence additions. Each method used baseline/no-RAG, structured/no-RAG, baseline/RAG, and structured/RAG, yielding eight conditions. Baseline prompts requested the distributions directly. Structured prompts added the following guidance, simplified here:

> For each requested configuration, consider 100 hypothetical comparable cases. Estimate the child given the parents, distinguish evidence from assumptions, and check that probabilities sum to one.

The hypothetical cases supplied no observations; fractional expected counts were allowed. RAG added qualitative evidence cards

to either prompt style. These additions changed elicitation support, not the requested rows or output format.

RAG supplied fixed, curated paraphrases of domain explanations. Retrieval matched cards to model context and target/parent tags, selecting up to three; packets contained 24–103 words (median 72). Cohesion received cards on shared goals, organizing work, and knowledge coverage. The first card read:

> **Shared goals.** Cohesive teamwork includes commitment to shared tasks and team goals. Agile collaboration involves joint responsibility, rather than treating every task as an isolated individual obligation.

The other cards linked adapting work and overlapping expertise to flexibility. None supplied conditional frequencies: probabilities remained subjective. No-RAG requests retained the definitions without cards. Packets were reused across methods, prompt styles, and replications, excluding reference probabilities, parameters, and identifying links. RAG tested qualitative context, not live retrieval or company measurements; input budgets were not equalized.

3.3 Producing comparable complete tables

Direct outputs were complete CPTs. RNM fitted weights and variance to the selected rule and anchor distributions, then generated all rows [11, 15]. It minimized mean anchor TV without accessing reference probabilities; valid poor fits were retained. Equations, numerical settings, and influence-direction evidence appear in the supplement.

Anchors comprised all-low, all-high, all-middle, one-high/rest-low, and one-low/rest-high configurations, with duplicates removed independently of the reference. For Cohesion, RNM requests five distributions: $(0,0)$, $(2,2)$, $(1,1)$, $(2,0)$, and $(0,2)$, where 0, 1, and 2 denote ordered states. Both methods produce nine rows, all compared against the reference, including fitted anchors.

3.4 Execution and reference isolation

Five replications were collected on 15–16 October 2026 using Codex CLI 0.162.1, requesting `gpt-6-astra` at `medium` effort. Each target–condition–replication used a fresh session; requests were shuffled within replications. Temperature and generation seed were unspecified. Experimental sessions could not access references, full papers, evaluator code, or authoring conversations; tools, browsing, connectors, and memory were disabled.

Outputs required correct rows/states, finite probabilities in $[0,1]$, and sums within 10^{-6} of one. One fresh-session format repair was allowed without reference feedback. All 2,440 trials produced valid outputs after four repairs (two per method); accounting includes all 2,444 attempts, with no fallback distributions.

3.5 Outcomes and analysis

For reference row p and prediction q, lower total variation means closer agreement:

$$\mathrm{TV}(p,q) = \tfrac{1}{2}\sum_s |p_s - q_s| \in [0,1]. \quad (1)$$

We average TV over rows within each node, then give all 61 nodes equal weight, preventing larger CPTs from dominating. Within each replication, RQ1 averages RNM-minus-direct differences over four prompt/evidence combinations. RQ2 contrasts structured minus baseline and RAG minus no RAG within each method, averaging the other factor. All conditions use complete tables.

Scrum and Chrome supply 52 of 61 nodes, dominating the node-weighted mean. A sensitivity analysis gives each of the five publication families equal weight, averaging nodes within BNs and BNs within families. We also report each BN descriptively. Supplementary Jensen–Shannon divergence, ordinal-CDF distance, and smoothed CPT KL use the same weights.

RQ3 counts requested distributions before repairs and client-reported tokens over all attempts for the full sample. Total tokens equal input plus output; cached-input and reasoning-output counters are subsets, not additional tokens. Paired contrasts use replication totals. Latency and offline fitting time are descriptive; subscription tokens are not converted into monetary savings.

The five paired replication blocks are the inferential units, not correlated nodes or rows. Two-sided paired t-tests ($df = 4$) and pointwise 95% intervals assume approximately normal, independent, stationary block differences. Five blocks provide little evidence for these assumptions. Two-sided exact sign tests assess direction/median sensitivity under independence; five concordant nonzero differences still yield $p = 0.0625$. Holm adjustments cover two RQ1 weighting contrasts, eight RQ2 contrasts (two methods, two factors, two weighting rules), and two RQ3 token contrasts. These are separate exploratory families, not study-wide confirmatory control; intervals are not simultaneous. No prespecified practical-effect or equivalence margin supports claims of usefulness or equivalence.

3.6 Offline diagnostics

Supplementary diagnostics examine anchor fitting, alternative functions, numerical settings, and baselines. Reference-informed fits assess best-found approximation, not error floors. Historical non-anchor, Chrome-version, and node-omission analyses concern a 25-node subset; they do not replace the full-sample endpoint or predictions.

4 Results

We first compare the accuracy of the completed CPTs (RQ1), then ask whether prompting and RAG improve that accuracy (RQ2), and finally examine the effort saved by eliciting fewer rows (RQ3). All main results include every row of all 61 nodes. Lower TV means closer reference agreement; negative RNM-minus-direct differences favor RNM.

4.1 RQ1: Which method produces closer complete CPTs?

Direct elicitation is closer overall. Mean TV is 0.2317 for direct elicitation and 0.2404 for RNM, averaging each method's four prompt/evidence combinations and five replications. RNM increases error by 0.0087, or 3.76%. Figure 1b shows all five equal-node differences to the right of zero (0.0072–0.0094), indicating higher RNM error. The diamond and whiskers show the mean and its pointwise 95% t interval, $[0.0076, 0.0098]$. The Holm-adjusted paired-t test rejects ($p = 0.000048$), but the exact sign test does not ($p = 0.125$). The

Table 3: Complete-table TV by BN, averaged over prompt/evidence conditions and replications. Δ is RNM minus direct. Every node is included; BN rows are descriptive.

BN	Nodes	Direct	RNM	Δ
Scrum	25	0.2605	0.2334	−0.0272
Chrome 40	15	0.2069	0.2220	+0.0151
Chrome 42	12	0.1988	0.2335	+0.0347
APS	3	0.0964	0.1155	+0.0191
SESSI	1	0.1160	0.1143	−0.0017
Software logs 3	4	0.3802	0.4770	+0.0967
Software logs 4	1	0.2052	0.3299	+0.1247
All nodes	61	0.2317	0.2404	+0.0087

direction is consistent across runs, with limited distribution-free evidence; neither test establishes practical importance.

Scrum and Chrome supply 52 of 61 nodes. Figure 1b therefore also gives each source study equal weight. Direct elicitation still has lower TV: 0.1937 versus 0.2189, a difference of 0.0252 (12.99%), with interval $[0.0211, 0.0292]$, adjusted t $p = 0.000065$, and adjusted sign $p = 0.125$. Reducing the influence of larger studies retains the conclusion. Secondary complete-table distances agree under both weights (supplement).

The overall result masks unequal gains and losses. Table 3 gives each BN's mean errors; its Δ column and Figure 1a show their direction and magnitude. Scrum improves from 0.2605 to 0.2334, and SESSI improves slightly. Both Chrome versions, APS, and both log BNs worsen, with the largest increases in the five log nodes. RNM nevertheless wins on 31 of 61 individual nodes. Larger deteriorations outweigh those gains: the aggregate compares error magnitudes, not win counts. BN-level contrasts are descriptive, not seven separate population effects.

Two rows make the differences concrete. Figure 2 compares probability allocations for two configurations not queried by RNM. In panel (a), a Scrum team has Low Self-Organizing but Medium Collaboration. The reference assigns 76.6% probability to Low Cohesion. Direct elicitation assigns 58.0%; RNM assigns 67.6%, closer to the reference. Mean row TV falls from 0.1862 to 0.0901. Thus reconstruction can improve agreement beyond the elicited anchors.

Panel (b) shows a different outcome. A Chrome component has three or four contributors and no minor contributors. Contributors are classified by their share of changes, so all must be major: the Major Contributors node belongs in the three-to-four category. The reference and all direct responses assign probability one there. RNM assigns 84.4%, spreading 15.6% into neighboring categories. Its reconstruction weakens a count constraint preserved by direct elicitation. These selected examples explain opposing effects without establishing their prevalence across the sample.

Supplementary diagnostics on identical anchors in the 25-node compression subset show little change from fitting for Scrum (TV 0.2790 to 0.2762), but deterioration for Chrome (0.2491 to 0.2994). Alternative functions, searches, and reference-informed fits explore that contrast without establishing a single cause of the full-sample result. **RQ1 favors direct elicitation on average under both weighting rules, despite RNM gains in some nodes and BNs.**

Table 4: All eight conditions: complete-table TV across all 61 nodes, with equal node weights and five replications. Each row pairs the methods under the same prompt and evidence.

Prompt	Evidence	Direct	RNM	Δ
Baseline	None	0.2327	0.2396	+0.0070
Baseline	RAG	0.2349	0.2419	+0.0070
Structured	None	0.2282	0.2389	+0.0107
Structured	RAG	0.2311	0.2413	+0.0102

4.2 RQ2: Do richer prompts or qualitative evidence help?

Structured prompting offers a limited benefit. Table 4 pairs the methods under identical support. Read across rows to compare direct and RNM, and down each method's column to compare prompts and evidence. Every Δ is positive: direct elicitation remains closer in all four pairs. Its lowest mean TV, 0.2282, occurs with structured prompts and no RAG.

Averaging over evidence conditions, structured prompting lowers direct TV by 0.00413 (interval $[-0.00595, -0.00232]$, Holm-adjusted t $p = 0.0255$). This is the only RQ2 contrast that rejects after eight-test correction. Equal-family weighting gives a similar direction (−0.00508) but an interval crossing zero and adjusted $p = 0.4953$. No RQ2 exact sign test rejects. For RNM, the prompt effect is only −0.00067 and does not reject after adjustment.

Qualitative RAG provides no demonstrated gain. Adding RAG raises mean error under both prompt styles in Table 4. Averaged over styles, the change is +0.00252 for direct elicitation and +0.00230 for RNM; neither adjusted test rejects. The Cohesion card illustrates the information provided: shared goals and joint responsibility, without conditional frequencies. **RQ2 supports structured direct elicitation only under node weighting and demonstrates no benefit from this qualitative RAG.** This does not establish equivalence or rule out benefits from local quantitative evidence. Full contrasts appear in the supplement.

4.3 RQ3: Do fewer judgments translate into fewer tokens?

Elicitation shrinks more than token usage. RNM requests 5,040 distributions versus 8,420 for direct elicitation: 40.1% fewer. Both still make one initial request per node and condition, carrying shared instructions, definitions, and evidence. Table 5 separates generated output from the whole exchange. Output tokens fall by 12.12%, but total tokens fall by only 0.61%, because inputs dominate (approximately 14.36 million direct and 14.31 million RNM).

Across five paired replication totals, the adjusted t test supports output savings ($p = 0.000073$), but not total-token savings ($p = 0.0809$); both adjusted exact sign tests give $p = 0.125$. The supplement reports block differences and intervals. Fewer requested probabilities therefore do not produce proportional token savings.

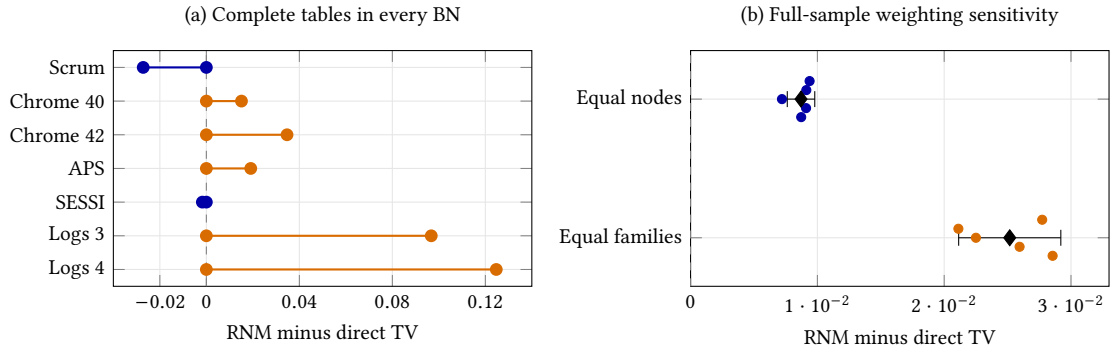


Figure 1: Complete-CPT agreement across the full sample. Negative differences favor RNM. (a) Each BN's mean averages all rows within nodes, then nodes, four prompt/evidence combinations, and five replications equally; points are descriptive. (b) Both weighting rules retain all 61 nodes. Circles show the five block differences; diamonds and whiskers show means and pointwise 95% t intervals. Equal-family weighting first averages nodes within BNs and BNs within each of five publication families.

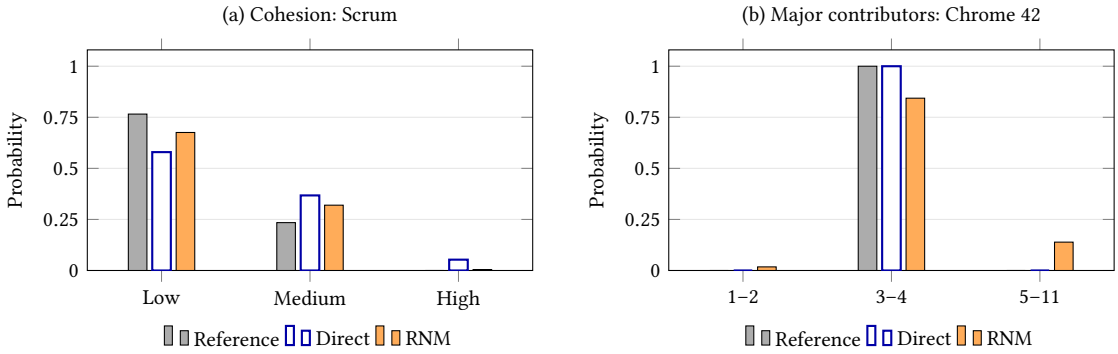


Figure 2: Two CPT rows unqueried by RNM. (a) Cohesion given Low Self-Organizing and Medium Collaboration. (b) Major contributors given three or four total contributors and no minor contributors. Bars average 20 predictions per method. Examples were selected for opposing effects in the earlier non-anchor comparison; they illustrate mechanisms and do not represent the full sample.

Table 5: RQ3: tokens across all 61 nodes and five replications, including repairs. Holm-adjusted paired-t tests compare five method-paired block totals across two counters; neither exact sign test rejects.

Counter	Direct	RNM	Reduction	Holm t p
Output	341,153	299,798	12.12%	0.000073
Input + output	14,697,090	14,607,517	0.61%	0.080921

The savings do not establish a better overall trade-off. Summed request latency is 3.82% lower for RNM; the original retained fits sum to 182.3 seconds of recorded fitting time, excluding subsequent offline diagnostics. These are computational descriptions, not controlled wall-clock or monetary savings. **RQ3 supports reduced elicitation and output size, with limited evidence of total-token savings.** Those savings accompany higher mean complete-table error across the sample, so they do not establish a superior accuracy–effort tradeoff.

5 Discussion

The three RQs distinguish reference agreement, the effects of elicitation support, and token usage. Together, they inform the selection and validation of a CPT construction pipeline rather than identify a universally preferable method.

5.1 Research questions in relation to the literature

RQ1: Reconstruction is an assumption, not an accuracy guarantee. RNM imposes a shared mathematical relationship across rows [11,

15]. This can stabilize uncertain judgments, but also spread error when the intended relationship differs. Scrum's RNM-derived reference makes compatibility plausible; Chrome's contributor-count example exposes a constraint that smoothing weakens. Restricted diagnostics support compatibility as an explanation, without isolating it from influence assumptions, LLM judgments, or optimization. The other BNs show why this interpretation cannot be reduced to a Scrum–Chrome dichotomy.

Blomaard et al. fit semi-automatic methods against known CPTs [2], examining what their representations can approximate. Our fits use uncertain LLM judgments, testing the whole pipeline. Nafar et al.'s gains from finer probability-query granularity [18] concern asking smaller questions, rather than omitting configurations. These comparisons distinguish elicitation quality from reconstruction compatibility: ordinal states make RNM applicable, but do not establish its suitability for every relationship.

RQ2: Instructions and evidence address different problems. Structured prompts clarify conditioning and normalization without supplying missing domain knowledge. Their limited benefit for direct elicitation is consistent with that distinction, although our design does not isolate which instruction helped. For RNM, improved judgments must also survive fitting. Weighting-sensitive findings do not establish a general prompting advantage across SE contexts.

Qualitative RAG reinforces meanings and relationships already partly expressed in the common brief; it cannot uniquely determine conditional frequencies. Gottal and Matthes instead use numerical summaries to constrain reconstruction [12], while Cypko et al. assess clinical modeling support [6]. Our result concerns numerical agreement under a particular evidence design, not RAG's value for modeling assistance. Whether company measurements improve elicitation remains untested.

RQ3: Human burden and token usage are different resources. Partial elicitation reduces the judgments experts must supply [1]. In an LLM workflow, repeated definitions and instructions can dominate the tokens used to return probabilities. Output compression therefore need not yield comparable total-token savings. Alternative batching or context reuse would require a new controlled comparison that also checks accuracy; our token totals establish neither monetary nor computational savings.

5.2 Implications for SE researchers and practitioners

For researchers. Evaluate complete CPTs alongside diagnostics separating raw judgments from fitting effects. Report aggregate error and variation across BNs: conclusions can depend on study representation or large deteriorations despite many small gains. Match the endpoint to the claim. BIRD evaluates LLM judgments coupled with BN inference [9]; similarly, SE studies should test whether CPT differences change release, prioritization, or process-improvement decisions. Reference agreement alone cannot demonstrate decision value or explainability.

For practitioners. Use direct elicitation as the initial numerical baseline suggested by this sample; consider RNM when its compensatory or constraining relationships match domain knowledge. Check logical constraints before accepting smooth distributions: contributor counts can exclude otherwise plausible states. Treat either output as a candidate requiring review. Generic definitions and reusable BN components can initialize a model, but organization-specific probabilities require local evidence and validation. Experts could focus on consequential or uncertain relationships, supported by sensitivity analysis and held-out company cases. This is a proposed validation strategy, not evidence that limited expert input necessarily suffices.

6 Threats to Validity

Construct validity. TV measures contextual CPT agreement, not observed calibration, expert replacement, explanation quality, or decision utility. Equal-row scoring includes potentially rare configurations. Scrum's reconstruction lacks an independently certified AgenaRisk export, and its RNM origin favors the tested representation. The seven-BN breakdown and restricted fitting diagnostics expose this distinction. SESSI represents an original elicitation stage. Adapted variable meanings and uncertain parent directions may affect agreement despite correct indexing.

Internal validity. Fresh sessions, shared definitions, reference isolation, and blinded fitting limit direct access leakage. Public benchmark memorization remains possible. The logs identify the requested model configuration, but cannot independently verify the exact serving-model version. Prompt and fitting development used the Scrum setting, which may favor that reference. Offline variants cannot exhaust all explanations. Reference-informed selection is optimistic in-sample diagnosis; bounded search may miss better fits. Schema validation cannot establish semantic or evidence fidelity, and exhaustive human review was not performed. The elicitation instructions and evidence cards were tested as bundles, so the contribution of each individual instruction or passage is unknown.

Conclusion validity. Five blocks give little evidence about the distribution or stationarity of differences. The paired-t and exact-sign results must be considered together; neither treats shared nodes or rows as independent replications. The SE restriction, completetable primary endpoint, weighting choices, and statistical plan were selected after inspecting earlier results; inference is therefore exploratory. Separate Holm families do not provide study-wide confirmatory error control. No prespecified acceptable-error margin supports claims of equivalence or practical sufficiency. Repeated execution estimates here are conditional on these fixed references and settings.

External validity. The 61-node sample spans seven BNs but only five publication families; Scrum and Chrome supply 52 nodes. Equal-family sensitivity changes their influence without making the sample representative. Some nodes offer no reduction in elicited rows, and tables have at most 27 rows. Only one requested model and reasoning effort were evaluated. Organizational distributions, larger CPTs, other model families, and evidence containing local numerical measurements may yield different tradeoffs.

7 Conclusion

We compared direct LLM elicitation with LLM-assisted RNM on complete CPTs for 61 ordinal nodes from seven SE BNs, including

anchors. Eight conditions varied method, prompting, and qualitative RAG over five replications.

Direct elicitation achieved lower mean TV: 0.2317 versus 0.2404, a 3.76% increase with RNM. Equal publication-family weighting retained the direction and increased the difference. RNM nevertheless improved 31 of 61 nodes and the mean for Scrum and SESSI; larger deteriorations elsewhere outweighed those gains. Reconstruction therefore cannot be recommended from ordinal eligibility or reduced input size alone. Both overall weighting contrasts passed Holm-adjusted paired t-tests, but not exact sign tests with five replications, limiting the strength of the statistical evidence.

Structured prompting provided a modest benefit for node-weighted direct elicitation, without consistent statistical support across weighting rules. Qualitative RAG demonstrated no benefit. RNM requested 40.1% fewer distributions and used 12.1% fewer output tokens, yet only 0.61% fewer total tokens. Its savings in returned information thus accompanied higher mean complete-table error, rather than establishing a superior accuracy–effort tradeoff.

Researchers should report complete-pipeline comparisons, weighting sensitivity, and network variation. For practitioners, LLM-generated CPTs remain candidate initializations requiring validation of probabilities and constraints in the intended organization. Future work should test local quantitative evidence, additional model configurations, larger CPTs, and prediction or decision quality on held-out projects. This would connect reference agreement to explainable SE decision support.

Replication Materials

The replication package contains the elicitation materials, reference CPTs, saved responses, fitted tables, evaluation scripts, token counts, and supplementary analyses. Reference materials are separated from collection inputs. The repository is private for author review.

Use of Generative AI

LLM judgments are experimental data (Section 3). Generative AI assisted drafting and script preparation/checking; numerical claims trace to saved responses and deterministic analyses.

References

[1] Martine J. Barons, Steven Mascaro, and Anca M. Hanea. 2022. Balancing the Elicitation Burden and the Richness of Expert Input When Quantifying Discrete Bayesian Networks. *Risk Analysis* 42, 6 (2022), 1196–1234. doi:10.1111/risa.13772

[2] Bodille P. M. Blomaard, Gabriela F. Nane, and Anca M. Hanea. 2025. Approaches for Reducing Expert Burden in Bayesian Network Parameterization. *Entropy* 27, 6, Article 579 (2025), 27 pages. doi:10.3390/e27060579

[3] Sicong Cao, Xiaobing Sun, Ratnadira Widyasari, David Lo, Xiaoxue Wu, Lili Bo, Jiale Zhang, Bin Li, Wei Liu, Di Wu, and Yixin Chen. 2025. A Systematic Literature Review on Explainability for ML/DL-based Software Engineering. *Comput. Surveys* 58, 4, Article 95 (2025), 34 pages. doi:10.1145/3763230

[4] Alexander Capstick, Rahul Krishnan, and Payam Barnaghi. 2025. AutoElicit: Using Large Language Models for Expert Prior Elicitation in Predictive Modelling. In *Proceedings of the 42nd International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 267)*. PMLR, 6746–6777. https://proceedings.mlr.press/v267/capstick25a.html

[5] Kwai-Sang Chin, Da-Wei Tang, Jian-Bo Yang, Shui Yee Wong, and Hongwei Wang. 2009. Assessing new product development project risk by Bayesian network with a systematic probability generation methodology. *Expert Systems with Applications* 36, 6 (2009), 9879–9890. doi:10.1016/j.eswa.2009.02.019

[6] Mario A. Cypko, Muhammad Agus Salim, Aditya Kumar, Leonard Berliner, Andreas Dietz, Matthaeus Stoehr, and Oliver Amft. 2026. Large language models with retrieval-augmented generation enhance expert modelling of Bayesian network for clinical decision support. *International Journal of Computer Assisted Radiology and Surgery* 21, 2 (2026), 211–222. doi:10.1007/s11548-025-03524-9

[7] Artur d'Avila Garcez and Luis C. Lamb. 2023. Neurosymbolic AI: the 3rd wave. *Artificial Intelligence Review* 56, 11 (2023), 12387–12406. doi:10.1007/s10462-023-10448-w

[8] Santiago del Rey, Silverio Martínez-Fernández, and Antonio Salmerón. 2023. Bayesian Network analysis of software logs for data-driven software maintenance. *IET Software* 17, 3 (2023), 268–286. doi:10.1049/sfw2.12121

[9] Yu Feng, Ben Zhou, Weidong Lin, and Dan Roth. 2025. BIRD: A Trustworthy Bayesian Inference Framework for Large Language Models. In *The Thirteenth International Conference on Learning Representations*. https://openreview.net/forum?id=fAAaT826Vv

[10] Norman Fenton, Martin Neil, William Marsh, Peter Hearty, Łukasz Radliński, and Paul Krause. 2008. On the effectiveness of early life cycle defect prediction with Bayesian Nets. *Empirical Software Engineering* 13 (2008), 499–537. doi:10.1007/s10664-008-9072-x

[11] Norman E. Fenton, Martin Neil, and Jose Galan Caballero. 2007. Using Ranked Nodes to Model Qualitative Judgments in Bayesian Networks. *IEEE Transactions on Knowledge and Data Engineering* 19, 10 (2007), 1420–1432. doi:10.1109/TKDE.2007.1073

[12] Jonas Gottal and Florian Matthes. 2026. Verifiable Parameterization of Bayesian Networks from Scientific Literature: Unlocking Unstructured Empirical Evidence. In *Findings of the Association for Computational Linguistics: ACL 2026*. Association for Computational Linguistics, 3400–3414. doi:10.18653/v1/2026.findings-acl.167

[13] Jan-Phillip Herrmann, Sven Tackenberg, Elio Padoano, Jörg Hartlief, Jens Rautenstengel, Christine Loeser, and Jörg Böhme. 2022. An ERP Data Quality Assessment Framework for the Implementation of an APS system using Bayesian Networks. *Procedia Computer Science* 200 (2022), 194–204. doi:10.1016/j.procs.2022.01.218

[14] Andrey Krutauz, Tapajit Dey, Peter C. Rigby, and Audris Mockus. 2020. Do code review measures explain the incidence of post-release defects? Case study replications and Bayesian networks. *Empirical Software Engineering* 25 (2020), 3323–3356. doi:10.1007/s10664-020-09837-4

[15] Pekka Laitila and Kai Virtanen. 2020. On Theoretical Principle and Practical Applicability of Ranked Nodes Method for Constructing Conditional Probability Tables of Bayesian Networks. *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 50, 5 (2020), 1943–1955. doi:10.1109/TSMC.2018.2792058

[16] Martí Manzano, Claudia Ayala, Cristina Gómez, Antonin Abherve, Xavier Franch, and Emilia Mendes. 2021. A Method to Estimate Software Strategic Indicators in Software Development: An Industrial Application. *Information and Software Technology* 129 (2021), 106433. doi:10.1016/j.infsof.2020.106433

[17] Emilia Mendes. 2012. Using knowledge elicitation to improve Web effort estimation: Lessons from six industrial case studies. In *Proceedings of the 34th International Conference on Software Engineering*. IEEE, 1112–1121. doi:10.1109/ICSE.2012.6227108

[18] Aliakbar Nafar, Kristen Brent Venable, Zijun Cui, and Parisa Kordjamshidi. 2026. Extracting Probabilistic Knowledge from Large Language Models for Bayesian Network Parameterization. *Transactions on Machine Learning Research* (May 2026). https://openreview.net/forum?id=Fy3Byg3CVo

[19] Mirko Perkusich, Kyller Gorgonio, Hyggo Almeida, and Angelo Perkusich. 2017. A Framework to Build Bayesian Networks to Assess Scrum-based Software Development Methods. In *Proceedings of the 29th International Conference on Software Engineering and Knowledge Engineering*. 7 pages. doi:10.18293/SEKE2017-139

[20] Mirko Perkusich, João Nunes, Emilia Mendes, Emanuel Dantas, Ademar Sousa, Danyllo Albuquerque, Kyller C. Gorgônio, and Angelo Perkusich. 2026. Semi-Automatic Quantification of Bayesian Networks for Software Decision Support: Comparing WSA and RNM in a Software R&D Organization. *arXiv preprint arXiv:2609.31275* (2026), 37 pages. https://arxiv.org/abs/2609.31275

[21] Mirko Perkusich, Gustavo Soares, Hyggo Almeida, and Angelo Perkusich. 2015. A procedure to detect problems of processes in software development projects using Bayesian networks. *Expert Systems with Applications* 42, 1 (2015), 437–450. doi:10.1016/j.eswa.2014.08.015

[22] Kumar Rahul and Shovan Chowdhury. 2026. Human–AI Construction of Bayesian Networks for Operational Decision Support: A Virtual Survey Approach. *arXiv preprint arXiv:2607.14141* (2026). doi:10.48550/arXiv.2607.14141

[23] Thiago Rique, Mirko Perkusich, Emanuel Dantas, Danyllo Albuquerque, Kyller Gorgónio, Hyggo Almeida, and Angelo Perkusich. 2023. On Adopting Software Analytics for Managerial Decision-Making: A Practitioner's Perspective. *IEEE Access* 11 (2023), 73145–73163. doi:10.1109/ACCESS.2023.3294823

[24] Thiago Rique, Mirko Perkusich, Kyller Gorgónio, Hyggo Almeida, and Angelo Perkusich. 2025. Constructing the graphical structure of expert-based Bayesian networks in the context of software engineering: A systematic mapping study. *Information and Software Technology* 177 (2025), 107586. doi:10.1016/j.infsof.2024.107586